

## 1. Discrete Distributions

|        | Bern(p)          | Binom(n,p)                   | Geom(p)                   | Pois( $\lambda$ )                   | Disc Unif<br>$\{1, \dots, n\}$    |
|--------|------------------|------------------------------|---------------------------|-------------------------------------|-----------------------------------|
| Range  | $\{0,1\}$        | $\{0,1,\dots,n\}$            | $\{1,2,3,\dots\}$         | $\{0,1,2,\dots\}$                   | $\{1,\dots,n\}$                   |
| PMF    | $p^x(1-p)^{1-x}$ | $\binom{n}{k}p^k(1-p)^{n-k}$ | $(1-p)^{k-1}p$            | $e^{-\lambda} \frac{\lambda^k}{k!}$ | $\frac{1}{n}$                     |
| $E[X]$ | p                | np                           | 1/p                       | $\lambda$                           | $\frac{n+1}{2}$                   |
| Var    | $p(1-p)$         | $np(1-p)$                    | $(1-p)/p^2$               | $\lambda$                           | $\frac{n^2-1}{12}$                |
| MGF    | $(1-p) + pe^t$   | $[(1-p) + pe^t]^n$           | $\frac{pe^t}{1-(1-p)e^t}$ | $e^{\lambda(e^t-1)}$                | $\frac{1}{n} \sum_{k=1}^n e^{tk}$ |

Binom = sum of n iid Bern(p) | Indicator:  $I_A \sim \text{Bern}(P(A))$  | Geom: memoryless (discrete) |  $E[X] = \sum x \cdot P(X=x)$  | LOTUS:  $E[g(X)] = \sum g(x)P(X=x)$  MGF domains: Geom  $t < -\ln(1-p)$  | Bern/Binom/Pois/Disc Unif: all  $t$  | General disc unif on  $\{x_1, \dots, x_n\}$ :  $E[X] = \frac{1}{n} \sum x_i$

## 2. Variance & Linearity

$E[\sum a_i X_i] = \sum a_i E[X_i]$  |  $E[aX+b] = aE[X] + b$   $\text{Var}(X) = E[X^2] - (E[X])^2$  |  $E[X^2] = \text{Var}(X) + (E[X])^2$   $\text{Var}(aX+b) = a^2 \text{Var}(X)$   $\text{Var}(\sum a_i X_i) = \sum a_i^2 \text{Var}(X_i)$  **only when independent** | E is always linear, Var is not

## 3. Continuous RVs

$f(x) \geq 0$ ,  $\int f(x)dx = 1$  (can be more than 1) |  $P(X=a) = 0$   $F(x) = \int_{-\infty}^x f(t)dt = P(X \leq x)$  |  $f(x) = F'(x)$  |  $P(a < X < b) = F(b) - F(a)$  | endpoints don't matter | Normalizing:  $f(x) = c \cdot h(x)$ , set  $\int c \cdot h(x)dx = 1$  | LOTUS:  $E[g(X)] = \int g(x)f(x)dx$  Quantiles:  $P(X \leq q_\alpha) = \alpha$  | if invertible,  $q_\alpha = F^{-1}(\alpha)$  | Median =  $q_{1/2}$  | IQR =  $q_{3/4} - q_{1/4}$  Nonneg RVs:  $E[X] = \int_0^\infty P(X > x)dx = \int_0^\infty [1 - F(x)]dx$  (other  $E[X]$  way)

## 4. Continuous Distributions

**Uniform**( $a, b$ ): Range  $[a, b]$  |  $f = \frac{1}{b-a}$  |  $F = \frac{x-a}{b-a}$  |  $E = \frac{a+b}{2}$  |  $\text{Var} = \frac{(b-a)^2}{12}$

**Normal**( $\mu, \sigma^2$ ): Range  $\mathbb{R}$  |  $f = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-\mu)^2}{2\sigma^2})$  |  $E = \mu$  |  $\text{Var} = \sigma^2$   $Z = (X-\mu)/\sigma \sim N(0,1)$  |  $P(a < X < b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$  |  $c = \mu + \sigma\Phi^{-1}(\alpha)$   $aX+b \sim N(a\mu+b, a^2\sigma^2)$  (linear transform stays Normal) | 68-95-99.7 rule

**Exp**( $\beta$ ): Range  $[0, \infty)$  |  $f = \beta e^{-\beta x}$  |  $F = 1 - e^{-\beta x}$  |  $P(X > x) = e^{-\beta x}$  |  $E = 1/\beta$  |  $\text{Var} = 1/\beta^2$   $q_\alpha = -\frac{1}{\beta} \ln(1-\alpha)$  | Memoryless:  $P(X > t+s | X > t) = P(X > s)$  | Exp = Gamma(1,  $\beta$ )

**Gamma**( $\alpha, \beta$ ): Range  $(0, \infty)$  |  $f = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$  |  $E = \alpha/\beta$  |  $\text{Var} = \alpha/\beta^2$  |  $\alpha$  = shape,  $\beta$  = rate

**Beta**( $a, b$ ): Range  $(0, 1)$  |  $f = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}$  | **Gamma fn**:  $\Gamma(a+1) = a\Gamma(a)$  |  $\Gamma(n) = (n-1)!$  |  $\Gamma(1/2) = \sqrt{\pi}$

## 5. Kernel recognition

$\int_0^\infty x^{\alpha-1} e^{-\beta x} dx = \frac{\Gamma(\alpha)}{\beta^\alpha}$  (gamma kernel: power =  $\alpha - 1$ , rate =  $\beta$ )  $\int_{-\infty}^\infty \exp(-\frac{(x-\mu)^2}{2\sigma^2}) dx = \sqrt{2\pi\sigma^2}$   
 (Gaussian kernel: complete the square) Complete square:  $-ax^2 + bx = -a(x - \frac{b}{2a})^2 + \frac{b^2}{4a}$

## 6. MGFs

$M_X(t) = E[e^{tX}]$  |  $M(0) = 1$  |  $M'(0) = E[X]$  |  $M''(0) = E[X^2]$  |  $\text{Var} = M''(0) - (M'(0))^2$  Same  
 MGF = same dist |  $M_{aX+b}(t) = e^{bt} M_X(at)$  | Indep:  $M_{X+Y} = M_X \cdot M_Y$  Domains: Exp  $M(t) = \frac{\beta}{\beta-t}$ ,  
 $t < \beta$  | Gamma  $M(t) = (\frac{\beta}{\beta-t})^\alpha$ ,  $t < \beta$  | Normal/Pois: all  $t$

## 7. Transformations

**CDF method:** (1) Range( $Y$ ) (2)  $F_Y(y) = P(g(X) \leq y)$  (3) Solve for  $X$ , use  $F_X$  (4) Differentiate  
 Step 2: isolate  $X$  inside  $F_X$ : e.g.  $P(2X + 3 \leq y) = P(X \leq \frac{y-3}{2}) = F_X(\frac{y-3}{2})$ . Then differentiate with  
 chain rule. Increasing  $g$ : inequality same | Decreasing  $g$ : flips (e.g.  $P(1/X \leq y) = P(X \geq 1/y) =$   
 $1 - F_X(1/y)$ ) **Monotone shortcut:**  $f_Y = f_X(g^{-1}(y)) \cdot |(g^{-1})'(y)|$  Non-monotone ( $Y = X^2$ ): use  
 CDF method,  $P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y})$ , simplifies if  $X > 0$

## 8. Joint Distributions

Discrete joint: same formulas, but **sums replace integrals** |  $p_X(x) = \sum_y p_{XY}(x, y)$  |  $p_{X|Y}(x|y) =$   
 $p_{XY}(x, y)/p_Y(y)$  |  $f_X(x) = \int f_{XY}(x, y) dy$  (marginal) |  $f_{X|Y}(x|y) = f_{XY}(x, y)/f_Y(y)$  (conditional)|  
 $f_{XY} = f_{Y|X} \cdot f_X = f_{X|Y} \cdot f_Y$  (decomposition) |  $f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)}$  (Bayes)| Independence:  
 $f_{XY} = f_X \cdot f_Y$  for all  $(x, y) \Leftrightarrow$  conditional = marginal | Hierarchical: given  $X \sim P_X$ ,  $Y|X = x \sim P_{Y|X}$ ,  
 joint =  $f_{Y|X}(y|x) \cdot f_X(x)$  |  $E[X|Y = y] = \int x f_{X|Y}(x|y) dx$  (function of  $y$ ) |  $E[g(X, Y)] = \iint g \cdot f_{XY} dx dy$   
 If indep:  $E[g_1(X)g_2(Y)] = E[g_1(X)] \cdot E[g_2(Y)]$  | **SKETCH JOINT RANGE FIRST** |  $0 < x <$   
 $y < \infty$ : fix  $y \rightarrow x \in [0, y]$ ; fix  $x \rightarrow y \in [x, \infty)$  |  $0 < y < x < 1$ : fix  $x \rightarrow y \in [0, x]$ ; fix  $y \rightarrow x \in [y, 1]$

## 9. Sums/Averages of IID RVs

$X_i$  iid, mean  $\mu$ , var  $\sigma^2$ , mgf  $M(t)$  |  $S_n = \sum X_i$  |  $\bar{X}_n = S_n/n$   $S_n$ :  $E = n\mu$  |  $\text{Var} = n\sigma^2$  | MGF  
 $= M(t)^n$   $\bar{X}_n$ :  $E = \mu$  |  $\text{Var} = \sigma^2/n$  | MGF =  $M(t/n)^n$  Bern  $\rightarrow$  Binom( $n, p$ ) | Pois  $\rightarrow$  Pois( $n\lambda$ ) |  
 Exp  $\rightarrow$  Gamma( $n, \beta$ ) Gamma( $\alpha, \beta$ )  $\rightarrow$  Gamma( $n\alpha, \beta$ ) |  $N(\mu, \sigma^2) \rightarrow N(n\mu, n\sigma^2)$ , avg  $N(\mu, \sigma^2/n)$

## 10. Reference & Tricks

$n = 1$  raw = mean |  $n = 2$  central = var |  $n = 3$  std = skewness |  $n = 4$  std = kurtosis From  $E[X^n]$   
 Skew: 0 symm,  $> 0$  right tail,  $< 0$  left |  $\sum r^k = 1/(1-r)$  |  $e^x = \sum x^k/k!$  |  $(x+y)^n = \sum \binom{n}{k} x^k y^{n-k}$   
 $\sum_{k=1}^n k = \frac{n(n+1)}{2}$  |  $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$  | use for disc unif mgf  $\rightarrow$  mean/variance Conditioning:  
 $P(X = k|A) = \frac{P(X=k, A)}{P(A)}$  | **Key pattern:** counting RV = sum of indicators  $\rightarrow$  use linearity for E,  
 independence for Var **MGF exists = all moments finite** (not converse) | **same MGF = same**  
**dist** (same moments does NOT)